1. Sum of Two Numbers

Write a function that takes two numbers as input from the user and prints their sum.

```
#WAP to sum two numbers

a = int(input("Enter first number: "))

b = int(input("Enter second number: "))

print("Sum= ",a+b)
```

output: Enter first number: 12

Enter second number: 8

Sum= 20

2. Area of a Circle

Function to input radius and calculate the area of a circle.

```
#WAP to calculate area of circle

radius = 12

print("Area of Circle: ", 3.14*radius*radius)
```

output: Area of Circle: 452.15999999999997

3. Simple Interest Calculator

Take Principal, Rate, and Time as input and calculate Simple Interest.

```
#WAP to calculate simple interest

principal = float(input("Enter principal Amount: "))

rate = float(input("Enter rate of interest: "))

time = float(input("Enter time period in years: "))

print("Simple Interest: ", (principal*rate*time)/100)
```

output: Enter principal Amount: 10000

Enter rate of interest: 10

Enter time period in years: 2

Simple Interest: 2000.0

4. Maximum of Two Numbers

Take two numbers as input and use a function to print the greater one.

```
#WAP to display maximum of two numbers
a = int(input("Enter first Number: "))
b = int(input("Enter second Number: "))
if a>b:
   print("A is Greater than B")
else:
   print("B is Greater than A")
```

output: Enter first Number: 12

Enter second Number: 10

A is Greater than B

Enter first Number: 10

Enter second Number: 12

B is Greater than A

5. Check Even or Odd

Function to check whether an input number is even or odd.

```
#WAP to print even or odd
a = int(input("Enter value: "))
if a%2 == 0:
      print("It is Even")
else:
   print("It is odd")
```

output: Enter value: 12

It is Even

Enter value: 3

It is odd

Level 2 – Moderate Difficulty

```
#WAP to find Factorial of a Number
a = int(input("Enter Number: "))
factorial = 1
for i in range(1,a+1):
   factorial *= i
   print(factorial)
```

output: Enter Number: 5

1

2

6

24

120

```
#WAP to check leaf year
year = int(input("Enter year: "))
if (year % 4 == 0 and year % 100 !=0) or (year % 400 == 0):
   print("It is Leap Year")
else:
   print("It is not Leap Year")
```

output: Enter year: 2025

It is not Leap Year

Enter year: 2020

It is Leap Year

```
#WAP to analyze student marks
def student_marks():
name = input("Enter student name: ")
m1 = float(input("Enter marks in subject 1: "))
m2 = float(input("Enter marks in subject 2: "))
m3 = float(input("Enter marks in subject 3: "))
total = m1 + m2 + m3
avg = total / 3
print("\nName:", name)
print("Total Marks:", total)
print("Average:", round(avg, 2))
if avg >= 75:
print("Grade: A")
elif avg >= 60:
print("Grade: B")
elif avg >= 50:
print("Grade: C")
else:
print("Grade: Fail")

# 9. Temperature Converter
def temp_converter():
c = float(input("Enter temperature in Celsius: "))
f = (c * 9 / 5) + 32
print("Fahrenheit =", round(f, 2))
f = float(input("Enter temperature in Fahrenheit: "))
c = (f - 32) * 5 / 9
print("Celsius =", round(c, 2))
```

10. Simple Calculator

def simple_calculator():

a = float(input("Enter first number: "))

b = float(input("Enter second number: "))

op = input("Enter operator (+, -, *, /): ")

if op == '+':

print("Result =", a + b)

elif op == '-':

print("Result =", a - b)

elif op == '*':

print("Result =", a * b)

elif op == '/':

print("Result =", a / b)

else:

print("Invalid operator")

Level 4 – Challenge Zone

11. Simple Voting Eligibility System

def voting_eligibility():

age = int(input("Enter your age: "))

if age >= 18:

print("You are eligible to vote.")

else:

print("You are not eligible to vote.")

12. Bill Calculator with Discount

def bill_calculator():

amount = float(input("Enter total amount: "))

```
if amount > 1000:

discount = amount * 0.1

else:

discount = 0

final = amount - discount

print("Discount =", discount)

print("Final Amount =", final)

# 13. Grade System using Average Marks

def grade_system():

marks = []

for i in range(5):

m = float(input(f"Enter marks of subject {i+1}: "))

marks.append(m)

avg = sum(marks) / 5

print("Average =", avg)

if avg >= 75:

print("Grade: A")

elif avg >= 60:

print("Grade: B")

elif avg >= 50:

print("Grade: C")

else:

print("Grade: Fail")

# 14. Menu-Driven Function Program

def menu_program():

print("1. Addition")

print("2. Subtraction")

print("3. Multiplication")
```

```
print("4. Division")
ch = int(input("Enter your choice: "))
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
if ch == 1:
print("Result =", a + b)
elif ch == 2:
print("Result =", a - b)
elif ch == 3:
print("Result =", a * b)
elif ch == 4:
print("Result =", a / b)
else:
print("Invalid choice")
# 15. Password Strength Checker
def password_checker():
pwd = input("Enter password: ")
has_upper = any(ch.isupper() for ch in pwd)
has_lower = any(ch.islower() for ch in pwd)
has_digit = any(ch.isdigit() for ch in pwd)
if len(pwd) >= 8 and has_upper and has_lower and has_digit:
print("Strong Password")
else:
print("Weak Password")
```